

Siyu Xie

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EDUCATION

The University of Chicago

Master of Science in Financial Mathematics

Chicago, IL

Expected December 2027

- **Courses:** Probability and Stochastic Processes, Options, Portfolio and Risk Management, Python

University of Nottingham

Ningbo, China & Nottingham, UK

Bachelor of Science in Computer Science with Artificial Intelligence (2+2; GPA: 3.9/4.0)

June 2026

- **Courses:** Calculus, Linear Algebra, Linear and Discrete Optimization, Discrete Mathematics, Machine Learning, Artificial Intelligence Algorithms, Designing Intelligent Agents, C++ Programming, Data Structures
- **Awards:** Gold Medal, WorldQuant Global Alpha Competition; Meritorious Winner (Top 6%), Mathematical Contest in Modeling (MCM)

SKILLS

Computing: Python, C++, SQL, PyTorch, Pandas, NumPy, Scikit-learn, Jupyter, Git, Linux

Knowledge: Factor Models, Backtesting, Risk & Portfolio Analytics, Optimization, Sequential Decision-Making

EXPERIENCE

Yingling Asset Management Company

Ningbo, China

Quantitative Researcher Intern

July 2025 - August 2025

- Built HFT risk-control components combining RL-based adaptive hedging, LSTM microstructure anomaly detection, and CNE5/Barra-inspired factor risk attribution across volatility, CVaR, drawdown, and correlation
- Developed a rolling forecasting and backtesting framework for a multi-strategy FoF model, achieving ~18% annualized out-of-sample return with a Sharpe ratio of 1.25

Ningbo Bank

Ningbo, China

Architecture Intern

July 2024 - September 2024

- Fine-tuned Qwen1.5-14B with Firefly and built a financial RAG system using LlamaIndex, improving Q&A accuracy by 7.6% and supporting automated generation of 50+ monthly investment analysis reports

RESEARCH

University of Nottingham

Nottingham, UK

Reinforcement Learning Order Book Trading Agent

March 2026 - June 2026

- Constructed a PPO-based trading agent in a BSE simulated limit order book with ZIC/ZIP background traders, comparing 5 state representations across price-only, best bid/ask, raw LOB depth, handcrafted, and encoded inputs
- Designed risk-sensitive rewards incorporating PnL, inventory risk, and transaction costs; handcrafted microstructure states achieved the highest mean PnL and outperformed non-learning baselines ($p < 0.0001$)

Graph-Enhanced Reinforcement Learning Hyper-Heuristics for VRPTW

October 2025 - April 2026

- Designed a GraphSAGE-PPO hyper-heuristic for vehicle routing with time windows, selecting among 6 operators based on graph-based features to improve solution quality and cross-instance generalization

University of Nottingham Ningbo China

Ningbo, China

Algorithm-Driven Quantitative Factor Mining

February 2025 - August 2025

- Built a modular factor research system to standardize RPN-based factor parsing, automated factor computation, IC/ICIR evaluation, regime-aware backtesting, and strict no-lookahead validation
- Integrated a Q-learning-based factor weighting module, improving cumulative reward from -10.07 to +0.58 and Sharpe from 0.459 to 0.536 on the Hull Tactical dataset while analyzing transaction-cost drag and regime sensitivity

Real-Time Trading Strategy System

June 2024 - November 2024

- Developed a PPO-LSTM trading system with 24-step rolling market states and multi-source sentiment signals to model short-term market dynamics, achieving a Sharpe ratio of 2.69 on the DOGE test set under transaction costs
- Co-authored an IEEE CIFEr 2025 paper on cryptocurrency trading strategies using deep reinforcement learning and multi-source LLM sentiment analysis; shortlisted for Best Paper Award

EXTRACURRICULAR

President, Computer Enthusiasts Association

September 2023 - September 2024

- Organized technical workshops, alumni Q&A sessions, and programming competitions for student members